+86 18925193410 | lw3227@columbia.edu | www.linxiaowu.com | github.com/lw3227

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, GPA 4.0/4.0 2026 12

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, GPA 3.7/4.0 2025 6

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CPU IP 2026 6 – 2026 9

- RVA23 Profile RISC-V CPU IP VPU (VFPU)
- VFCVT FP64/FP32/FP16/BF16 RVV
- **PPA** 3 Lane DLEN
- VFPU VCTRL vector uop mask/lane valid/stall/flush pipeline forwarding
- Synopsys Design Compiler RTL

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RD TCAD 2024 7 – 2024 9

- FAB 32V PMOS TCAD IGBT
- WAT

Apple **Systolic-Array-Based CNN** **7** 2026 2 –

- INT8 CNN ASIC 64×64 CNN
- 64 PE INT8 systolic-array 464k MAC 8KB SRAM TSMC 65nm
- IC RTL DFT scan chain Cadence Genus Innovus APR
- Python / MATLAB CNN

P6 **Out-of-Order RISC-V** 2026 3 -2026 6

- P6 RISC-V OoO 2-wide Fetch/Dispatch/Retire
- 16-entry ROB 8-entry Reservation Station Map Table ROB Tag CDB
- 5-bit GHR + 32-entry BHT/BTB Gshare 2-wide speculative GHR ROB snapshot branch misprediction
- Split LQ/SQ 2-way Non-blocking D-cache Store-to-Load Forwarding 4-entry MSHR 4-entry Victim Cache

64-tap 16-bit FIR 2025 10 - 2025 12

- 64-tap 16-bit FIR CMEM/IMEM FSM
- FIFO 10 kHz FIR CDC

EDA: Synopsys(Design Compiler, PrimeTime), Cadence(Genus, Innovus, Xcelium), JasperGold, Modelsim, Verdi
: Verilog, SystemVerilog, Python (NumPy, PyTorch), C, MATLAB›
ASIC RISC-V CPU/VPU CNN